

# A Semantic Fidelity Lexicon

## A Working Vocabulary for Meaning Preservation Across AI and Representational Systems

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### Abstract

This lexicon defines the core vocabulary of the Semantic Fidelity Project and places each term within the Reality Drift framework. The terms distinguish conditions of representational integrity, mechanisms that alter meaning, failures within AI workflows, optimization dynamics, and ecosystem effects. The vocabulary is intended for diagnosis. It identifies what changed, how the change accumulated, and which form of corrective contact weakened.

### Core Claim

A useful vocabulary must separate the quality being preserved, the process of degradation, the mechanisms that produce it, and the system-level condition that allows it to persist.

### Foundational Conditions

**Semantic Fidelity:** The degree to which meaning is preserved as information moves across systems, media, or representations. It describes the integrity of the relationship between a representation and the reality, source, intent, or referent it is meant to preserve.

**Semantic Continuity:** The persistence of recognizable meaning across successive transformations. Continuity indicates that later representations remain interpretable as descendants of the original, but it does not by itself prove high fidelity.

**Semantic Drift:** The gradual degradation of meaning as information passes through compression, transformation, interpretation, and reuse. Representations remain coherent and recognizable while their relationship to the original meaning weakens.

**Reality Drift:** The system-level condition in which representations progressively displace the realities they were created to represent. The system remains operational and internally coherent while losing corrective contact with external conditions.

**Fidelity Decay:** The cumulative reduction of semantic fidelity across repeated transformations. Fidelity decay describes the trajectory of loss rather than a separate mechanism.

## Mechanisms of Loss

**Recursive Compression:** A process in which systems reduce complexity into simplified representations and then reuse those representations as inputs for later interpretation and compression. It enables learning and coordination but creates drift when fidelity is not preserved.

**Recursive Mediation:** The growing dependence of interpretation on prior layers of representation. Each mediator encounters an already processed account rather than the original condition.

**Context Loss:** The removal of conditions needed to interpret a representation correctly, including scope, history, audience, uncertainty, and situational purpose.

**Hierarchy Flattening:** The loss of relative importance among claims, evidence, exceptions, and examples. Information remains present but no longer carries its original weight.

**Constraint Collapse:** A condition in which feedback, consequence, and correction remain present but lose their ability to enforce change. The system can register error without being altered by it.

**Ground Erosion:** The progressive loss of accessible contact with the evidence, experience, or conditions from which a representation arose.

## Failures in AI Workflows

**Retrieval Drift:** A change in meaning introduced when selection and ranking produce a source set that is individually relevant but structurally unrepresentative of the available evidence.

**Interpretation Failure:** A transformation that preserves source material while assigning it the wrong role, implication, or significance.

**Memory Drift:** The accumulation of altered meaning as compressed interaction history is stored and reused to shape later output.

**Agent Drift:** The gradual divergence of an agent's behavior from the user's intended objective as intermediate representations, plans, and success criteria become substitutes for the original task.

**Verification Failure:** A failure to test a representation against the evidence or condition capable of correcting it.

**Verification Substitution:** The replacement of contact with the underlying condition by confirmation that a procedure, citation, review, or formal check occurred.

## Optimization Dynamics

**Proxy Optimization:** The process by which a system improves a representation of an outcome rather than the outcome itself. The proxy begins as a guide and becomes the effective target.

**Proxy Substitution:** The point at which a metric, category, or procedure begins functioning as the reality it was intended to track.

**Evaluation Blindness:** The inability of an evaluation system to detect losses outside the properties it measures. An output can improve against the evaluation while becoming less faithful to its referent.

**Synthetic Coherence:** The appearance of completeness and internal consistency produced through generative reconstruction. It can make missing evidence, context, or causal structure difficult to see.

**Optimization Trap:** A condition in which continued improvement against an internal objective deepens misalignment because the objective no longer preserves the underlying purpose.

## Ecosystem Effects

**Recursive Synthetic Feedback:** A loop in which generated outputs become inputs for later generation, training, evaluation, or human interpretation.

**Provenance Failure:** The loss of a recoverable path between a representation and its source. Agreement among derivative outputs can then be mistaken for independent confirmation.

**Citation Laundering:** The transfer of apparent authority through citation chains that repeatedly refer to derivative accounts while obscuring the original evidentiary basis.

**Evidence Laundering:** The transformation of uncertain, partial, or context-bound material into apparently settled evidence through repetition and institutional reuse.

**Synthetic Realness:** The felt plausibility of a representation produced through coherence, familiarity, and repetition rather than sustained contact with the represented condition.

**Lexical Decay:** The weakening of a term's discriminating meaning as it is repeatedly expanded, generalized, optimized, or used ceremonially.

## The Drift Principle

The Drift Principle describes the pressure that ties these terms together. Representations can be compressed, optimized, distributed, and reused faster than reality can correct them. As that happens, systems lean more heavily on prior representations, selected versions gain authority, and the connection to the referent can weaken. If correction no longer has enough force to interrupt the process, that loss can persist and accumulate. Reality Drift is the broader condition that emerges from that pattern.

## Using the Vocabulary

Diagnosis begins by identifying what is being represented, what it is meant to remain connected to, and which parts of the meaning matter most. From there, the task is to trace where the loss entered, what rewarded or preserved it, and why correction failed to restore contact.

The terms in this lexicon describe different parts of that process. Semantic fidelity concerns the strength of the connection itself. Semantic drift describes its weakening. Recursive compression and related mechanisms show how loss can accumulate, while constraint collapse explains why that loss can continue without being corrected. Reality Drift names the broader condition that can emerge when those failures persist.

## Conclusion

The Semantic Fidelity Project is concerned with what happens to meaning as representations are transformed, reused, and passed through increasingly complex systems. The vocabulary is meant to help locate loss before it becomes obvious as failure.

Coherence is not contact.